DHAIRYA PRAJAPATI

Software Engineer | AI & Machine Learning | Backend Development | Data & Quantitative Analysis

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dhairya19793.github.io | Available Summer 2027 | CPT-Eligible F-1 Student

EDUCATION

California Polytechnic State University, San Luis Obispo Aug 2026 - May 2030

B.S. Computer Science | First-Year Computer Science Student

Coursework: AP Calculus AB, AP Computer Science Principles | Planned Spring 2027: Data Structures (CSC 2001), Calculus II | Harvard CS50x: Introduction to Computer Science | Pre-College (CBSE, India/UAE): Computer Science, Mathematics, Physics, Chemistry, English

EXPERIENCE

AI in Healthcare Research Intern | *Pharmint Pte Ltd | Remote (Singapore)* Jun 2026 - Jul 2026

- Completed a 6-week internship in **predictive modeling and biomarker discovery** in cancer transcriptomics, mentored by a computational biologist.
- Engineered **differential gene expression analysis** (log2FC, adjusted p-values) and trained ML classifiers (Logistic Regression, Random Forest) to identify hypoxia-responsive gene signatures in breast cancer transcriptomics data (TCGA-BRCA, GEO).

Research Author | *Researchory | Remote Research Program* Jun 2025

- Published a research paper on Researchory about **household energy analytics**, applying exploratory data analysis (EDA) to identify usage patterns and efficiency opportunities.
- Documented methodology and data-driven efficiency recommendations for publication, strengthening **technical writing and data storytelling** skills.

Research Mentee, Quantitative Analysis | *CSPIT, CHARUSAT | Remote (India)* Mar 2025 - Apr 2025

- Conducted **stock trend analysis** in Python (pandas, statsmodels) using regression, moving averages and time series methods under faculty mentorship.
- Presented findings and visualizations to a faculty mentor, summarizing trend patterns and **forecasting insights** for further discussion.

Data Science Simulation, Customer Analytics | *British Airways (Forage) | Remote* Mar 2025

- Completed a virtual job simulation, applying **web scraping and customer profiling** in Python to segment customers for personalized marketing decisions.

Finance & Banking Intern | *Grey Sells | Dubai, UAE* Jul 2024

- Shadowed banking professionals to learn **corporate finance, KYC, and valuation** processes over a two-week internship.
- Assisted with basic **financial modelling** exercises, building foundational understanding of valuation techniques.

Python Intern | *BizzAppDev Systems Pvt. Ltd. | Remote (India)* Jul 2023 - Aug 2023

- Contributed **backend modules** to a live Python/PostgreSQL/XML codebase, following modular design and coding standards.
- Debugged and maintained backend modules using Git-based **version control** on Linux, supporting documentation and software maintenance.

PROJECTS

StudyBuddy, Cal Poly CodeBox Hackathon | *Flask · Supabase · Live Demo: studybuddy-calpoly.bolt.host* Sep 2026

- Built a full-stack **study coordination platform** that matches students into private Study Pods by course, topic, availability and study style, owning the matching logic in Flask and the real-time, privacy-protected data layer in Supabase.
- Presented at the Cal Poly CodeBox Hackathon. Shipped a deployed web app spanning Course Hubs, Study Requests and a Solo Workspace for independent task tracking.

Jarvis, Offline AI Voice Assistant | *Python · Whisper · Ollama (Llama 3)* May 2026

- Built an offline voice assistant integrating speech recognition, a locally run Llama 3 model and text-to-speech, with **zero cloud dependency** for privacy and responsiveness.
- Iterated through multiple development and testing cycles for reliable, **privacy-preserving local performance**.

Study Planner, CS50x Capstone (HarvardX) | *Python · Flask · SQLite* Jul 2025

- Designed a full-stack study planning tool with relational data modeling, parameterized queries and **ACID-compliant transactions** for data integrity, built from concept through deployment.

CERTIFICATIONS

- **Harvard CS50x**: Introduction to Computer Science, HarvardX/edX (Verified) · Jul 2025.
- **Great Learning** (Verified): AI with Python, SQL for Data Science, Business Finance Foundations · Apr - May 2025.
- **Udemy** (Verified): Data Analytics & AI, Finance Modeling (NASBA CPE) · Mar 2025.
- **PurpleTutor**: Python Programming, Verified with Distinction · Jan 2022.
- **TDRA Virtual Academy** (Verified): Cloud Computing Services.

LEADERSHIP, COMMUNITY & HONORS

- **Honors**: SCO International Science Olympiad: Gold & International Outstanding Performance (2022-23), Rank 44/2,981 (2023-24) | International Olympiad Foundation: Top 42% in AI | askIITians iCAT: Country Rank 6 | CBSE Yogasana Nationals (U-19): Two-Time National Qualifier (2024, 2025), UAE Cluster Gold (2024) & Silver (2025).
- **Founder & Reviver, PIES Gavel Club**: Revived the school's public speaking club and organized regular meetings.
- **Delegate, PIESMUN**: Represented Madagascar's position in the SOCHUM Committee, applying research, problem-solving, communication and interpersonal skills.
- **Volunteer Tutor, Angel Xpress Foundation**: 40+ hours of Science & Math tutoring, plus 140+ verified community service hours overall.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript | Concepts: Object-Oriented Programming, Data Structures, Algorithms
Frameworks & Backend: Flask, Supabase, SQLite, MySQL, PostgreSQL, REST APIs, relational data modeling
AI / ML: pandas, NumPy, scikit-learn, statsmodels, EDA, regression, classification, time series forecasting
Quantitative & Finance: Financial Modeling, Valuation, Statistical Analysis, Business Finance Foundations
Tools & Platforms: Git, GitHub, Linux, Excel, Tableau (basic), R (basic), Whisper, Ollama